

How It Works

Physics Formula Quick Search Tool — Technical Explanation

Students:	Amina Fida Areeba Adeel
Department:	Department of Physics and Applied Mathematics
University:	Pakistan Institute of Engineering and Applied Sciences
Course:	AICT
Instructor:	Sir Atif Raza
Lab Assistant:	Sir Mosa
Submission:	May 3, 2026

Step 1 — The Formula Database

All physics formulas are stored in a Python list called FORMULAS. Each formula is a dictionary with 5 keys: name, formula, topic, variables, and keywords. The keywords list makes searching fast — instead of searching the raw formula text, we match against simple lowercase keyword strings.

```
{ "name": "Ohm Law", "formula": "V = I x R", "topic": "Electricity", "variables":  
"V=voltage | I=current | R=resistance", "keywords":  
["ohm", "voltage", "current", "resistance"] }
```

Step 2 — The Search Function

When the user types in the search box, the search_formulas() function is called immediately. It takes two inputs: the search query string and the active topic filter. It converts the query to lowercase, loops through every formula, and adds a formula to results if the query matches any keyword, the formula name, or the equation text.

```
def search_formulas(query, topic_filter): query = query.lower().strip() results =  
[] for f in FORMULAS: if topic_filter != 'All' and f['topic'] != topic_filter:  
continue if query == '' or any(query in kw for kw in f['keywords']):  
results.append(f) return results
```

Step 3 — The GUI Window

The graphical interface is built using Python's built-in tkinter library. The entire app is written as a class called PhysicsSearchApp. When the program starts, it creates the main window and calls _build_ui() which builds all the visual elements on screen.

UI Element	What it does
Header Label	Shows app title and student/university information
Entry widget	The text box where the user types their search
trace('w',...)	Calls the search function automatically every time text changes
Filter Buttons	Each button sets the active topic and re-runs the search
Canvas + Frame	Creates a scrollable area to hold the formula result cards
_render_card()	Draws one colored card for each matching formula found

Step 4 — Rendering Formula Cards

For each formula in the search results, the `_render_card()` method is called. It creates a Frame widget styled as a card. On the left side is a thin colored bar whose color depends on the formula's topic. Inside the card: the formula name and topic tag on top, the equation in large bold text in the middle, and variable explanations at the bottom.

Topic	Color Used
Mechanics	Orange
Energy	Yellow
Waves	Blue
Electricity	Purple
Thermodynamics	Red
Gravitation	Teal
Circular Motion	Green

Step 5 — Live Search with Event Binding

The search works in real-time because of tkinter's StringVar and trace mechanism. The search box is linked to a StringVar object. Using `trace('w', callback)`, Python automatically calls our search function every single time the text in the box changes — no button press needed.

Step 6 — Mac Scroll Fix

On macOS, the MouseWheel event sends delta values differently than Windows. On Windows, delta is a multiple of 120. On Mac, delta is already in scroll units. The scroll handler uses `int(-1 * event.delta)` directly, which correctly scrolls up and down on macOS 11 Big Sur.

Program Flow Summary

Step	What Happens
1	Program starts -> root window created -> PhysicsSearchApp() runs
2	<code>_build_ui()</code> draws header, info block, search box, filter buttons, scroll canvas
3	All 30+ formulas shown on startup via <code>_show_results(FORMULAS)</code>
4	User types a keyword -> trace fires -> <code>search_formulas()</code> called automatically
5	Matching formulas returned -> old cards cleared -> new cards drawn on screen
6	User clicks a topic button -> topic filter set -> search re-runs instantly
7	User clicks Clear -> <code>search_var</code> set to empty string -> all formulas shown again